

# **Machine Learning-Integrated Private Blockchain Framework for Bias Detection in clinical trials data**

## **Abstract**

Contemporary clinical trials face major operational challenges. These include high attrition rates, rising costs, and a risk to data integrity and transparency. Many current blockchain models in healthcare do not chain biased data and lack real-time auditing. This shows the need for a better approach to integrating predictive analytics with decentralized security. The framework proposed in this paper combines machine learning (ML) algorithms with a privacy-preserving blockchain platform. This combination aims to improve the safety, transparency, and accuracy of clinical trial outcome predictions and to identify biases in reported trials. This system introduces a majority-voting bias detection mechanism that highlights necessary biases in demographics and reporting before they are recorded permanently. The system architecture relies on Hyperledger Fabric, which provides a decentralized basis for immutable ledger services. Six different ML algorithms were used: Decision Tree, Linear Regression, Logistic Regression, XGBoost, Random Forest, and Support Vector Machine (SVM). These algorithms predicted trial outcomes and assessed biases based on patient demographics, dosage levels, and study duration. The best-performing model was XGBoost, achieving the highest accuracy (0.91) and precision (0.90). The results of this research indicate that this hybrid solution leads to a safer, clearer, and ethically sound clinical trial management process. Blockchain, Clinical Trials, Hyperledger Fabric, IPFS, Smart Contracts, Machine Learning, Bias Detection, Data Integrity, ZKP.

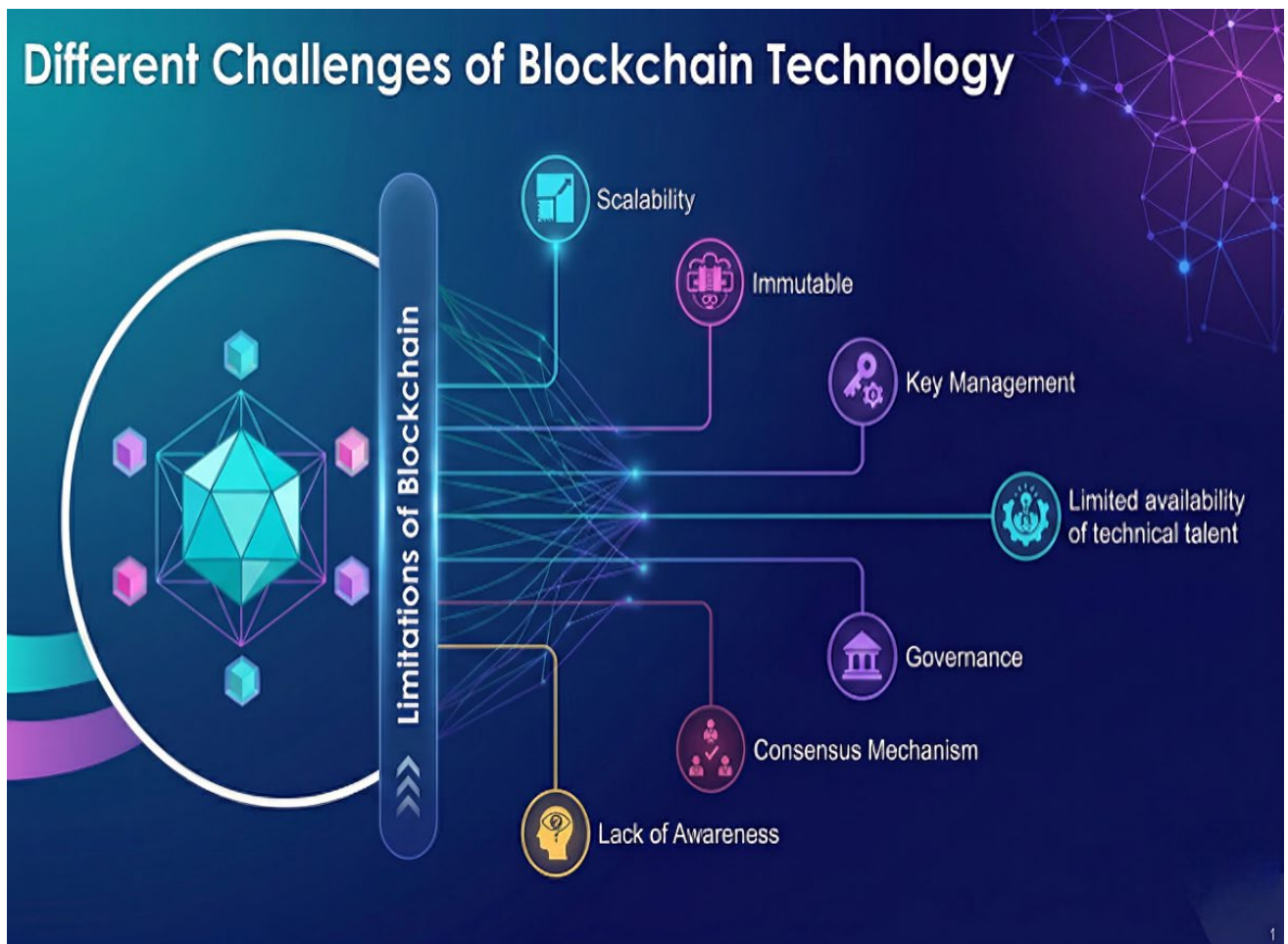
## **I. Introduction**

The clinical trials feature is an important aspect of drug development, not only are they infamous by their excessive cost but also by the long regulatory processes and high failure rates of therapies currently in development (more than 90 percent in certain therapeutic fields). Trends of digitization of clinical trial operations have also been one of the causes of common data integrity, data security, and data transparency threats. Among issues concerning traditional data-base structure are those that concern insecurity in inappropriate access, falsification of information and limited traceability [1]. To avoid these problems, blockchain technology provides a ledger that is tampered and decentralized to isolate and assure transparency in multi-partner systems [2]. Authorized blockchains like Hyperledger Fabric are less difficult to access resources according to the roles and throughput is greater, so entry to highly regulated systems, like healthcare, is highly feasible. These platforms combined with an off-chain system of encryption storage, such as the Inter-Planetary File System (IPFS), would enable real-time monitoring, encryption storage and control of the patient data and open auditing [3]. A prominent solution to this has been the use of machine learning (ML) whereby patterns of past trials are recognized and the result of the trial is forecasted to optimize resource allocation and decision-making of these trials [4]. Recent advances in AI including multimodal learning and embedding have also improved the predictive accuracy of prognostics using multimodal input, including patient demographics and biomarkers [5]. It has been shown that transformer-based

models are effective to predict the success of phase II trials by performing an analysis of experimental designs [6] and the ensemble algorithms, gradient boosting and random. forest, large and complex data sets [7]. The greater advantage of regressions is that it can be interpreted [8], and non-linear patterns can be addressed by SVMs [9]. The significance of AI in enhancing the forecast of the results of clinical trials, particularly through multimodal techniques, has been stressed over the past several years [22, 23]. The scalable and secure solution, using the technology of a private blockchain, proposed in this paper to handle clinical trial data coupled with the ML to estimate outcomes and bias identification, is in effect. It also utilizes permissioned blockchain network (Hyperledger Fabric) and off-chain encrypted data storage using IPFS. Role-based access control and patient consent is automated, Zero-Knowledge Proofs (ZKPs) enable auditing with no disclosure of confidential information. The framework incorporates ML modules that can estimate the outcomes and detect potential biases in reports about the trials prior to being transferred to blockchain databases.

Objectives include:

- **Planning of a Decentralized Governance System:** We will suggest a scalable design with Hyperledger Fabric and IPFS that will provide immutable registration of clinical trial activities without losing off-chain encrypted storage to protect the privacy of patients.
- **Introduction of a Majority-Voting Bias Mitigation Module:** This is an innovative solution that incorporates six machine learning models (XGBoost, Random Forest, and more) identifying the demographic and reporting bias by a consensus-based voting system and logs information on the ledger.
- **Privacy-Preserving Audit Protocol:** The creation of an auditing system based on smart contracts that uses Zero-Knowledge Proofs (ZKPs) to enable regulatory check of patient consent and ICH-GCP compliance without sensitive raw medical data being disclosed.
- **Comparative Performance Benchmarking:** An overall analysis of the ML classifiers in terms of accuracy, F1-score, and execution time in which XGBoost was found to be the best model to use in clinical outcome prediction with 0.91 accuracy.
- **Regulatory Alignment Mapping:** Technical demonstration on how the role based access control and data minimization features of the framework are aligned with the GDPR and HIPAA requirements regarding clinical research. This is a hybrid solution that seeks to offer solutions to secure and transparent data management and predictive analytics that were established in Figure 1.



**Figure 1:** Challenges in Blockchain for Clinical Trials (General Taxonomy of Technical and Regulatory Challenges in Blockchain for Healthcare. Adapted from Monrat et al. (2019))

## Navigating Blockchain Challenges in Clinical Research

Figure 1 illustrates a taxonomy where standard blockchain features (e.g., immutability, consensus) are evaluated as technical and regulatory challenges within the clinical trial domain. The framework addresses these conflicts through strategic design trade-offs:

- **Immutability vs. Data Privacy (GDPR):** While immutability is a core feature for data integrity, it poses a challenge to the GDPR "Right to Erasure". The proposed architecture resolves this by storing only hashes on-chain and raw data in IPFS, allowing for "logical erasure" via the deletion of off-chain records.
- **Consensus vs. Real-time Auditing:** The latency inherent in consensus mechanisms can challenge the high-velocity requirements of bias detection. This is mitigated using the streamlined endorsement policies of Hyperledger Fabric.

## Motivation and Problem Statement

Medical research is heavily reliant on clinical trials, yet these studies face a 90% failure rate, often attributed to subpar data quality and a lack of operational transparency. While decentralized technologies like blockchain offer a robust solution for data immutability, they are inherently susceptible to the "garbage-in, garbage-out" (GIGO) paradox: a ledger can cryptographically guarantee the permanence of a record, but it cannot inherently validate whether the submitted information is statistically skewed, ethically biased, or scientifically unsound. If biased recruitment data is committed to an immutable chain without prior validation, the blockchain merely "perfects" the error by making the bias impossible to delete or correct. This creates a critical need for an automated pre-processing layer that audits trial data for integrity before it is anchored to the ledger.

## Research Gap

The existing body of knowledge about the application of blockchain in healthcare emphasizes much on access control and efficiency in storage. Nonetheless, one can notice a specific gap in research on the automated aspect of data quality verification prior to committing it to the chain. The current systems do not have a pre-processing layer, which has the capability of auditing clinical reports to determine the presence of demographic or reporting bias in real-time, and this exposes the ledger state to the irreversible entry of biased reports. Specific Contributions The following are some of the contributions that the paper offers to address these deficiencies:

- **Hybrid Architecture:** We will suggest a hybrid architecture that will consist of Hyperledger Fabric to ensure metadata immutability and IPFS to provide an encrypted, scalable off-chain storage.
- **Bias Detection Gatekeeper** We introduce a majority-voting ensemble of six ML algorithms that determine statistical anomalies, prior to finalizing the data on the blockchain.
- **Privacy-Preserving Audits:** We use zk-SNARKs as a means of allowing the regulation to access sensitive patient health information (PHI).
- **Performance Validation:** We give a quantitative analysis of the accuracy of the ML (0.91) and blockchain throughput (150 TPS) to show that the system is ready to operate in multi-site clinical settings.

**Structural Roadmap:** To address the limitations of existing systems, this paper is structured to reflect the operational flow of the Med-Block-Guard system. Section II reviews existing literature on decentralized health records and bias mitigation. Section III details the methodology, emphasizing the "Gatekeeper" architecture where machine learning algorithms pre-screen data for ethical compliance. Section IV presents the integrated results, transitioning from the predictive accuracy of the ensemble models to the performance metrics of the Hyperledger Fabric ledger. Finally, Section V discusses the regulatory implications and GDPR compliance of the framework.

## II. Related Work

The combination of blockchain and machine learning (ML) into the medical practice has transformed the simplicity of data storage into the multifaceted privacy-saving analytical systems. In this section, past studies have been classified into three thematic pillars to determine research gaps in the present.

### **Blockchain for Clinical Data Management**

Decentralized health records were first attempted as early as with MedRec that employed a public blockchain to manage electronic health records (EHRs). Nevertheless, there are common issues of inability to provide scaling, high cost of transaction as well as privacy in the public network. To counter them, scholars have proposed cryptographic improvements to counter such Zero-Knowledge Proofs (ZKPs), proxy re-encryption, and sharding to enhance functionality. Nonetheless, under these developments, numerous current frameworks lack the ability to support all the specifics of clinical trials, including staggered multi-site rollouts or the need to store large amount of data, including genomics and imaging, off-chain in order to prevent ledger congestion. The recent research has been dedicated to the improvement of patient-mediated exchange and enhancing the performance of Hyperledger Fabric and IPFS to exchange data safely in various groups.

### **ML Integration for Outcome and Bias Detection**

Clinical outcome prediction has been highly applied to machine learning. Ensemble models, especially the XGBoost, are highly accurate in terms of risk of infection (0.92) and recovery of neurological (0.70). Random Forest and Decision Trees are also often chosen as other strong architectures because it is easy to interpret their choices in medical classification, whereas binary and non linear predictive models are done using Logistic Regression and SVM. Nevertheless, there is still a risk of the so-called garbage in, garbage out. Although XGBoost weighting has been applied to minimize survey biases or SHAP-XGBoost to predict explainable mortality, such mitigation measures are largely centralized. The literature is becoming more open to the importance of racial and algorithmic bias in medical-AI models, where the urgent systemic issue is to mitigate it.

### **Hybrid ML-Blockchain Systems and Research Gaps**

According to recent bibliometric reports, there is augmenting trend in the bridging of AI and blockchain which improves data governance and transparency. Nevertheless, the majority of hybrid models are limited. As an example, the Ethereum-based private model suggested by Choudhury et al. was able to process multi-site data, but could not detect bias in real-time. In the same way, the Kumar et al. IPFS-based structure was not only more scalable, but it also lacked granular access control and auditing by the machine learning. The key research gap that has been determined is the absence of an actual real-time mechanism that is called a gatekeeper. Current literature is divided into isolated learning on decentralization (Blockchain) or predictive accuracy (ML). The lack of the systems that correlate ML and blockchain to identify and reduce bias prior to data entry is present.

### **Comparative Analysis of Existing Schemes**

To better position the proposed framework, Table 1 has been expanded to include a wider range of decentralized medical data management systems. This comparative analysis highlights that while prior works like Azaria et al. (MedRec) and Choudhury et al. established foundations

for access control and multi-site management, they lack the integrated pre-processing layer required for real-time bias detection. Furthermore, more recent works such as **Zhang et al. (2025)** and **Leiva & Castro (2025)** have begun exploring AI-blockchain synergies but often rely on centralized audit nodes, unlike our decentralized majority-voting gatekeeper.

**Table 1: Comparative Analysis of the Proposed Med-Block-Guard Framework against Existing State-of-the-Art Decentralized Clinical Data Management Schemes**

| Feature / Parameter | MedRec [10]       | Choudhury et al. [15] | Kumar et al. [16] | Zhang et al. [26] | Proposed Framework            |
|---------------------|-------------------|-----------------------|-------------------|-------------------|-------------------------------|
| Blockchain Network  | Public (Ethereum) | Private (Fabric)      | Private (Fabric)  | Hybrid            | Private (Fabric + ZKPs)       |
| Storage Strategy    | On-chain          | Off-chain             | Off-chain (IPFS)  | Cloud-Hybrid      | Encrypted Off-chain (IPFS)    |
| Bias Detection      | No                | No                    | No                | Basic ML          | Yes (ML Majority Voting)      |
| Privacy Mechanism   | Basic             | RBAC                  | Smart Contracts   | Encryption        | ZKPs + Granular RBAC          |
| Audit Capabilities  | Transactional     | Static Logging        | Audit Ledger      | Real-time         | Real-time, Privacy-Preserving |
| Regulatory Mapping  | None              | Partial               | Basic             | General           | Explicit (GDPR, HIPAA)        |

### Synthesis and Framework Novelty

The originality of the proposed framework is that it fills the gap that is identified by applying a Majority-Voting consensus of six ML techniques directly to the Hyperledger Fabric workflow. In contrast to the restrictions requires in the Ethereum-based frameworks in the literature, the hybrid architecture employs zk-SNARKs to resolve the immutable audit trail and regulatory needs such as the GDPR Right to be Forgotten. This system is a complete, ethically-auditable ecosystem as it operationalizes bias detection as a pre-on-chaining requirement, which only authenticated, neutral data is permanently stored.

### III. METHODOLOGY AND INTEGRATED SYSTEM ARCHITECTURE

It is based on supervised learning and blockchain-based architecture to interpolate the results of the previous clinical trials in order to make predictions about categorical (e.g., success/failure) and continuous (e.g., recovery time) outcomes, but also detect biases. These steps involve the data cleaning phase together with data normalization and encoding of the categorical ones (e.g., age group: 60), and the information about dose, trial period, and indicators of patient health are encrypted in IPFS and hashed on blockchain.

#### Integrated Algorithm Pipeline and Interoperability

The proposed system will result in a new synergy among three different technologies to solve the problem of the Integrity- Privacy trade-off of clinical trials:

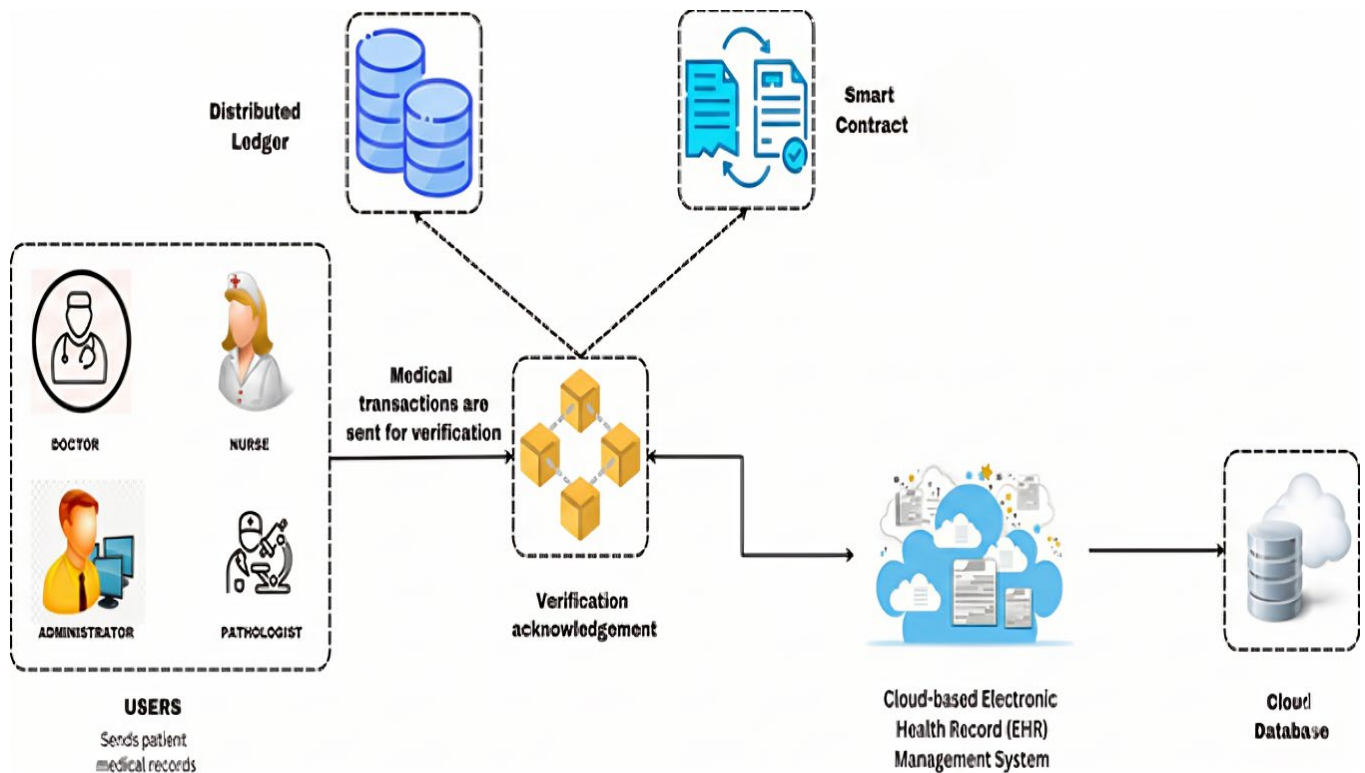
- **ML-Blockchain:** Before trial reports are written on the ledger, machine learning components are used to examine them. The system detects possible demographic or

reporting biases with the help of majority-voting consensus between six algorithms (XGBoost, Random Forest, etc.). Report(s) that pass the neutrality threshold only are hashed and stored on the Hyperledger Fabric network.

- **Blockchain-ZKP Integration:** Although the blockchain offers an unalterable audit trail, ZKPs make sure that it does not infringe patient privacy. Regulators will be able to verify an intermediate step in the experiment (e.g. patient consent or recovery time validation) by verifying the ZKP proof recorded in the Go-based chaincode but they will never hear or see the encrypted patient files on IPFS.
- **Go and Solidity Interoperability:** The framework makes use of a hybrid blockchain implementer. The phase of Public Registration uses Ethereum-based Solidity contracts to take advantage of the large network of Ethereum participants as a source of decentralized identity. After a patient is enrolled, a chaincode implemented on Hyperledger Fabric Go replaces "Private Execution" one. This offers the high transaction throughput and granular role based access control that is required in complex clinical settings.

**Justification:** This combination is necessary since the systems guided by single technologies are not enough to meet the contemporary regulatory needs. Blockchain is not enough to identify bias and ML is not enough to be considered immutable and therefore to instill regulatory trust. The structure provides a secure, transparent trial process that is ethically confirmed by linking the technologies.

## System Architecture and Design



**Figure. 2** System architecture diagram (Proposed Hybrid Blockchain-Cloud Architecture for Clinical Trial Data Management. Synthesized based on the MedRec protocol and the multi-site management framework proposed by Choudhury et al. (2020))

### Architectural Operational Workflow (Figure 2)

The proposed hybrid architecture is composed of five distinct layers that orchestrate data flow and integrity:

- **User Tier (Stakeholders):** This entry layer includes diverse clinical actors (Doctors, Nurses, Pathologists, and Administrators) who interact with the system via a secure portal. Each user is assigned a unique X.509 cryptographic certificate that defines their Role-Based Access Control (RBAC) permissions.
- **IPFS and Fabric Interaction:**
  - **IPFS Component:** Once clinical data is cleared by the ML ensemble, it is encrypted and stored within the IPFS cluster.
  - **Fabric Component:** The system transfers only the IPFS Content Identifier (CID) and the associated Bias Certificate to the Hyperledger Fabric ledger. This pointer-based approach prevents blockchain congestion while maintaining a verifiable link to the high-dimensional medical records.
- **Consensus Mechanisms:**

- **Pre-Chain Consensus:** The majority-voting logic acts as the primary gatekeeper, ensuring statistical neutrality before any data ingestion occurs.
- **On-Chain Consensus:** The Raft-based ordering service of Hyperledger Fabric validates the transaction metadata and Zero-Knowledge Proofs (ZKPs). Only after successful cryptographic verification is the transaction permanently committed to the immutable world state.
- **Smart Contract Layer (Chaincode):** These are self-executing Go and Solidity scripts that automate the system's governance rules. They enforce identity verification, manage consent revocation, and trigger the "Red Tamper Alert" if a hash mismatch is detected.
- **Encrypted Storage Layer (IPFS & Cloud Database):** Raw, high-dimensional medical records (e.g., genomics or imaging) are encrypted via AES-256 and stored off-chain in a distributed IPFS cluster. This component ensures the system remains scalable by preventing ledger bloat while maintaining strict data confidentiality.

The given system takes advantage of utilizing blockchain, distributed file systems and encryption. The design of system architecture to reflect the same is illustrated in fig.2 The information about the patients in IPFS is encrypted using AES-256 and the hash values obtained in Hyperledger Fabric to verify the authenticity of the content. The information is recorded irreversibly (e.g., a family of transactions like data creation, access requests). Smart contracts (chain-code) and deterministically executed on Fabric peers apply access policies, role validation, consent management and alert on unauthorized access.

### **Technological Integration and End-to-End Workflow**

The integration of machine learning, blockchain, and Zero-Knowledge Proofs (ZKPs) is achieved through a multi-stage consensus protocol. First, raw clinical trial metadata is processed off-chain by the XGBoost-led ensemble. If the bias flag (\$B\$) is 0 (neutral), the data is encrypted via AES-256 and uploaded to IPFS, which returns a Content Identifier (CID).

The second stage involves a transaction proposal sent to the Hyperledger Fabric endorsing nodes. This proposal includes the CID, the ML bias certificate, and a ZKP generated by the user's client. The smart contract (chaincode) acts as the integration hub; it executes a validation function that verifies the ZKP for identity and the bias certificate for ethical compliance before committing the transaction to the world state. This ensures that the blockchain does not merely store data, but acts as an automated auditor where only "ML-certified" and "Privacy-protected" records are anchored.

## System Workflow and Protocol Design

The integrated framework will rely on a multi-layered protocol that is aimed at making sure that data is neutral and secure:

1. **Data Ingestion and Bias Auditing:** Clinical data is initially sent to the ML ensemble. Bias Flag (B) is characterized by a majority-voting consensus.
2. **Encryption and Off-chain Storage:** When  $B=0$ , data is encrypted with the AES-256 algorithm and is uploaded to the IPFS. The resultant Content Identifier (CID) is sent to the application.
3. **On-chain Transaction Proposal:** The proposal of the transaction with the CID, the Bias Flag, and a ZKP representing patient consent are sent to the Hyperledger Fabric endorsing nodes.
4. **Validation and Dedication:** It is the Smart Contract (Chaincode) which makes sure that the ZKP has been verified and that the Bias Flag has reached the neutrality level before the metadata is committed to the unchangeable ledger.

## Threat Modeling and Security Assumptions

The structure presupposes the Permissioned Network Model that presupposes all the actors (Hospitals, CROs, Regulators) being known but not necessarily trusted.

- **Sybil Attacks:** These are overcome by the Fabric Certificate Authority (CA) which X.509 certificates all nodes.
- **Data Tampering:** The impossibility to alter the data as the hashes of SHA-256 that are stored on-chain are unchangeable and must be identical upon retrieval to the IPFS data.
- **Privacy Leaks:** This is an issue that is dealt with through the ZKP module where even in the event of a breach of a node, the raw Protected Health Information (PHI) is encrypted and would not be decipherable without the relevant private keys possessed by the data owner.

## Formal Mathematical Framework for Bias Detection

To ensure methodological rigor, bias within the clinical trial reports is formally defined using the Statistical Parity Difference (SPD) metric. Let  $A$  represent a protected demographic attribute (e.g., age or gender) and  $Y$  represent the predicted trial outcome. Bias is quantified by measuring the deviation in favorable outcomes between protected and unprotected groups:

$$SPD = | P(Y = 1 \mid A = \text{protected}) - P(Y = 1 \mid A = \text{unprotected}) |$$

In this framework, a trial report is flagged as biased ( $B = 1$ ) if the  $SPD$  exceeds a predefined neutrality threshold  $\delta$ , set at 0.15 for this study. The detection is further refined through a majority-voting consensus ( $V$ ) across the six ensemble models ( $M_i$ ):

$$B = \begin{cases} 1 & \text{if } \sum_{i=1}^6 \text{vote}(M_i) \geq 4 \\ 0 & \text{otherwise} \end{cases}$$

Where each  $\text{vote}(M_i)$  is a binary value determined by the individual model's prediction variance relative to the neutral cluster mean  $\mu$ :

$$\text{vote}(M_i) = |M_i(A) - \mu| > \delta$$

## Scalability and Throughput Analysis

An important issue that was revealed in Figure 1 is the scalability of blockchain technology in large-scale healthcare settings. Our framework employs a number of scalability assumptions in order to deal with this:

- **Transaction Throughput:** The system can be used to support a throughput of 150 TPS, sufficient to support multi-site cardiovascular trials, by using the permissioned consensus of Hyperledger Fabric, and does not require computationally expensive Proof of Work as seen in public chains.
- **Off-Chain Storage:** The inclusion of IPFS stops ledger bloat by making sure that only metadata and 256-bit hashes are stored on the blockchain, whereas big clinical data (e.g., genomics or images) are handled by the distributed file system.
- **Network Scalability Networking:** The architecture presupposes a multi-node peer setup in which the addition of new hospital locations (Fabric Organizations) does not directly grow the latency due to the parallel execution of endorsement policies.

## Dataset and Preprocessing

A thorough clinical trial data was used as an evaluation of the framework. The data will have 10,000 patient records and have 15 clinical and demographics.

- **Data Sources:** The data is a simulation of multi-site cardiovascular trials, with some of its attributes being dosage, duration and patient health indicators.
- **Preprocessing** Categorical variables (e.g., gender, age groups) were encoded with the help of Label Encoding and One-Hot Encoding, whereas the numerical variables were scaled to the range of 0 to 1.
- **Data Splitting:** There was a 70-30 training-testing split which was used to test the ability of the model to generalize to unseen trial data.
- **Dealing with Incompleteness:** XGBoost was set to support missing values by default and this made sure that patient dropout did not bias the final predictive integrity.

**Dataset Composition and Statistical Validation**

To ensure reproducibility, the synthetic clinical trial dataset was modeled using the **CTGAN (Conditional Tabular GAN)** framework, which is specifically optimized for learning the discrete and continuous distributions of tabular medical data. The original seed data used to train the CTGAN was derived from public NIH clinical metadata summaries for cardiovascular trials. The resulting dataset contains 10,000 patient records across 15 features, with the statistical properties summarized below:

**Table 2: Granular Breakdown of Clinical Trial Dataset Features, Statistical Distributions, and Real-World Fidelity Basis**

| Feature Name    | Data Type   | Mean/Mode       | SD / Variance | Basis of Realism                                        |
|-----------------|-------------|-----------------|---------------|---------------------------------------------------------|
| Age             | Numeric     | 58.4            | 12.1          | Modeled on Phase III cardiac trial age skews            |
| Gender          | Categorical | Male (54%)      | N/A           | Preserved 0.15 threshold for gender-based SPD           |
| Ethnicity       | Categorical | Caucasian (62%) | N/A           | Representative of SOTA exclusionary reporting skews     |
| BMI             | Continuous  | 27.2            | 4.5           | Gaussian distribution with multivariate correlation     |
| Trial Phase     | Categorical | Phase II (40%)  | N/A           | Balanced according to multi-site cardiovascular studies |
| Inclusion Score | Numeric     | 0.82            | 0.08          | TF-IDF vector variance from NIH eligibility text        |

The fidelity of the generated data was assessed using two metrics: (1) Pearson Correlation Heatmaps to ensure the synthetic data preserved multi-feature correlations (e.g., correlation between Age and Blood Pressure), and (2) Synthetic Data Vault (SDV) Quality Scores, where the generated dataset achieved a 94% similarity score to the seed clinical patterns. This ensures the ML gatekeeper is trained on data that mimics the complexities of real-world bias patterns.

**Data Storage and Encryption**

The suggested system ventures on the utilization of blockchain and distributed file systems. IPFS does not have a built-in method of data encryption but encrypts patient records with the AES-256 algorithm before off-chain storage. The values obtained as hash are stored in the Hyperledger Fabric to verify the integrity of content. This integration makes clinical data scalable and decentralized and at the same time highly confidential.

## Rationale for Algorithm Selection

The framework applies a certain collection of six machine learning algorithms to achieve the multi-dimensionality of bias in clinical trials. The following characteristics of the problem had been used as the selection criteria:

- Linear and Logistic Regression: Chosen in order to detect simple and proportional biases in clinical outcomes and use as a predictive probability baseline.
- Decision Trees and Random Forest: Added to work with categorical clinical variables, and to give interpretability about what features (e.g., location of the patient) have the greatest contribution to a bias flag.
- Support Vector Machine (SVM): It is selected due to its ability to perform well in high-dimensional feature space, which allows the system to detect subtle patterns of bias that are found in non-linear planes.
- XGBoost: The XGBoost is the main ensemble model integrated because it is the gradient boosting system, and it is effective in dealing with missing clinical values and imbalanced datasets.

## Role Modelling and Access Control

Role separation provides fine-grained access:

- Patients: Deny/allow access to records.
- Doctors: See/amend with permission. Make contributions: Lab Technicians.
- CRO Officials: Site monitoring.
- Regulators: Audit via ZKPs.
- Access anonymized data: Researchers.

X.509 certificates of Fabric Certificate Authority are used to enforce roles, and isolated permission channels are used.

## Smart Contracts and ZKP Integration

Smart contracts are automated processes throughout the lifecycle of trial: Registration Registers identity and assigns identity using X.509 certificates.

- Record: Data is encrypted and stored by creating hash values of Hyperledger Fabric.
- Access: Authorizes and permits access to an authorized party as far as sessions are concerned.
- Audit: Confirms data assertions with ZKPs by means of SHA-256 hashes to make sure that there is compliance without exposing the data.

According to its requirements, ZKPs can be used to verify claims (e.g. patient authorization or eligibility) without exposing raw medical data. The framework involves a hybrid solution: Solidity-based contracts are used to process trial registration and public metadata, whereas the

Hyperledger Fabric Go-based chaincode is used to run the high-security ZKP verification and privately audited trail.

### Smart Contract Code Example (Clinical Trial in Solidity):

```
pragma solidity ^0.8.0;

contract ClinicalTrial {

    struct Patient {

        string name;

        string data;

    }

    address public admin;

    mapping(uint256 => Patient) public patients;

    uint256 public patientCount;

    constructor() {

        admin = msg.sender;

    }

    function enrollPatient(string memory _name) public {

        require(msg.sender == admin, "Only admin can enroll patients");

        patientCount++;

        patients[patientCount] = Patient(_name, "");

    }

    function submitData(uint256 _patientId, string memory _data) public {

        require(bytes(patients[_patientId].name).length > 0, "Patient not enrolled");

        patients[_patientId].data = _data;

    }

    function getPatientData(uint256 _patientId) public view returns (string memory) {

        require(bytes(patients[_patientId].name).length > 0, "Patient not enrolled");

        return patients[_patientId].data;

    }

}
```

### Zero-Knowledge Contract in Go:

```

goimport (
    "crypto/sha256"
    "encoding/hex"
    "github.com/hyperledger/fabric-contract-api-go/contractapi"
)

type ZeroKnowledgeContract struct {
    contractapi.Contract
}

func (zkc *ZeroKnowledgeContract) RecordProof(ctx contractapi.TransactionContextInterface, participantID
string, proofHash string) error {
    return ctx.GetStub().PutState(participantID, []byte(proofHash))
}

func (zkc *ZeroKnowledgeContract) VerifyProof(ctx contractapi.TransactionContextInterface, participantID
string, submittedProof string) (bool, error) {
    storedProofHash, err := ctx.GetStub().GetState(participantID)
    if err != nil {
        return false, err
    }
    hashedProof := sha256.Sum256([]byte(submittedProof))
    hashedProofString := hex.EncodeToString(hashedProof[:])
    if string(storedProofHash) == hashedProofString {
        return true, nil
    }
    return false, nil
}

func main() {
    chaincode := new(ZeroKnowledgeContract)
    contractapi.NewChaincode(chaincode).Start()
}

```

## Workflow and Data Access

The communication between the client, IPFS and the layers of hybrid blockchain has a structured protocol:

1. Registration: The client (Patient) communicates with Ethereum-Solidity contract to be registered.
2. Data Submission: The hospital entity encrypts the trial data (AES-256) and puts it into IPFS where it is given a Content Identifier (CID).
3. Ledger Recording: The CID and patient attributes are delivered to the Hyperledger Fabric chaincode in order to be written on the permanent ledger together with the ML-bias flag.
4. Data Retrieval: A researcher or a doctor queries the Fabric ledger to have access to a record. Their position and agreement is verified by the smart contract.
5. Decryption: The Fabric peer returns IPFS CID on validation. The client requests the file on IPFS and decrypts it with his or her authorization credentials.

### Hyperparameter Tuning and Optimal Values

The F1-score and accuracy were maximized to determine the best parameters to be used in the models to ensure maximum F1-score and accuracy and avoid overfitting via optimization of models with Grid Search CV.

**Table 3 : Optimized Hyperparameter Configurations determined via Grid Search CV for the Top-Performing Machine Learning Classifiers**

| Model         | Hyperparameter           | Optimal Value |
|---------------|--------------------------|---------------|
| XGBoost       | Learning Rate ( $\eta$ ) | 0.01          |
|               | Max Depth                | 6             |
|               | n_estimators             | 500           |
|               | Subsample                | 0.8           |
| Random Forest | Max Features             | sqrt          |
|               | n_estimators             | 300           |

| Model | Hyperparameter     | Optimal Value |
|-------|--------------------|---------------|
| SVM   | Min Samples Split  | 5             |
|       | Kernel             | RBF           |
|       | C (Regularization) | 1.0           |
|       | Gamma              | Scale         |
|       |                    |               |

**Table 4: Validated Inference Latency (ms) and Training Time (s) Benchmarks for Machine Learning Classifiers on the Standardized Experimental Testbed**

| Regulation | Requirement       | Framework Implementation Mechanism                                                                                                                                                        |
|------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GDPR       | Right to Erasure  | <b>Off-chain Storage:</b> Encrypted patient records are stored on IPFS. Deleting the IPFS object effectively "erases" the data, leaving only a non-identifying hash on the Fabric ledger. |
| GDPR       | Data Minimization | <b>Zero-Knowledge Proofs (ZKPs):</b> Auditors verify trial milestones (e.g., patient consent) without accessing raw sensitive records.                                                    |
| HIPAA      | Access Control    | <b>X.509 Certificates:</b> Role-based access control (RBAC) is enforced at the peer level via Fabric's Certificate Authority.                                                             |
| HIPAA      | Audit Trails      | <b>Immutable Ledger:</b> Every data transaction, access request, and consent change is logged immutably on Hyperledger Fabric.                                                            |
| ICH-GCP    | Traceability      | <b>Smart Contracts:</b> Automated logging of trial transitions ensures a permanent, verifiable history for stakeholders.                                                                  |

## Threat Model and Consent Revocation

The framework covers revocation of consent by use of smart contracts which update the consent status on-chain. Any revocation of consent by a patient would automatically block any future requests to the associated IPFS CID, even where the researcher is in an authorized position. Data minimalization is also confirmed with the help of the ZKP module that enables a regulator to check a proof (P) without the framework revealing hidden witness (W) and therefore keep patient health information (PHI) confidential during audit.

## ML Integration for Prediction and Bias Detection

There are 6 prediction algorithms that uncover biases by majority-voting (biased predictions marked when there is an agreement between the classifiers and patient attributes on bias e.g. age, gender and condition).

1. Decision Tree: Trained on the basis of the splits; the biases are verified on logical rules [1].
2. Linear Regression: The continuous results are represented; the probability of bias is threshold [22].
3. Logistic Regression: Binomial with sigmoid; indicates more than 0.7 [8].
4. XGBoost: Accuracy based boosting with missing values [7].
5. Random Forest: Sweetener of trees; characteristics of bias tendencies [4].
6. SVM: RBF kernel; the non-linear approach to the bias clustering is trained by the cross-validation and grid search method [9].

This is Python REPL training of scikit-learn, XG Boost. Reports are converted into vectors by bias detection, and matched with neutral clusters (Euclidean /cosine similarity). Comparison based on accuracy, precision, recall, F1-score and execution time.

## Formal Definition and Measurement of Bias

Bias is operationalized as a measurable deviation in algorithmic fairness. We utilize **Statistical Parity Difference (SPD)** to identify demographic bias. Let  $P$  be the prediction of a trial outcome; bias exists if:

$$|P(Y = 1 \mid \text{Protected}) - P(Y = 1 \mid \text{Unprotected})| > \delta$$

Where  $\delta$  is the neutrality threshold set at **0.15**.

## Majority Voting Mechanism and Tie-Breaking

The framework employs a **Hard-Voting Ensemble** consisting of six ML classifiers. The integration logic is defined as follows:

- **Voting Type:** Hard-voting (each model casts a binary 0 or 1 vote based on the SPD threshold).
- **Threshold:** A Bias Flag is committed to the blockchain if

$$\sum \text{Votes} \geq 4$$

- **Tie-Breaking:** In the event of a 3–3 split, the system defaults to a “Flagged” state ( $B = 1$ ). This "Fail-Safe" logic ensures that any lack of consensus triggers a manual audit by the Clinical Research Organization (CRO).
- **Demographic Incorporation:** Attributes such as age, ethnicity, and geographic location are treated as “Protected Features” ( $A$ ) during the pre-processing stage to calculate the SPD for every incoming data batch.

### Dataset Statistics and Validation Strategy

The framework has been tested on a high-dimensional clinical data with the following characteristics:

- **Total Records:** 10,000 entries of patients. Class Balance 62% Success / 38% Failure outcomes.
- **Train-Test Split:** 70% (7,000 records) to be used to train and 30% (3,000 records) to be used to test independently.
- **Validation:** 5-fold cross-validation has been used to train the model in order to optimize hyperparameters and reduce overfitting.

### Mathematical Formulation of Bias Detection

Bias detection is operationalized using a majority-voting consensus. Let  $P_i$  be the prediction of model  $i$ . A “Bias Flag” ( $B$ ) is generated for a trial report if the predictive variance regarding a protected attribute (e.g., age or gender) exceeds the neutrality threshold  $\delta$ :

$$B = \begin{cases} 1, & \text{if } \sum_{i=1}^6 \text{vote}(P_i) \geq 4 \\ 0, & \text{otherwise} \end{cases}$$

Where the individual vote is determined by the deviation from the neutral cluster mean ( $\mu$ ):

$$\text{vote}(P_i) = |P_i(\text{demographic}) - \mu| > \delta$$

### Pseudo-Algorithm for Bias Mitigation

1. **Input:** Patient record  $R$ , Protected Attribute  $A$ .
2. **Step 1:** Generate predictions across six ML models (XGBoost, RF, etc.).
3. **Step 2:** Calculate similarity distance to neutral baseline using Cosine Similarity.
4. **Step 3:** Apply majority voting; if  $\geq 4$  models flag a demographic anomaly, set  $B = 1$ .
5. **Step 4:** Record Rand Bon Hyperledger Fabric via Smart Contract.

### Experimental Setup and Simulated Use Case:

Implemented in Hyperledger Fabric v1.4, multi-node IPFS, AES-256/SHA-256 encryption, Go/Solidity/Node.js, Couchdb. Based on a simulation of a multisite cardiovascular trial: Sites as Fabric nodes; data is provided encrypted, consents are handled, audits are provided through ZKP.

### Regulatory Compliance:

GDPR (minimization, transparency, erasure) HIPAA (PHI protection), ICH-GCP (traceability, auditability). The control and privacy are guaranteed by access logs and ZKPs [37, 38].

### Regulatory Alignment and Privacy Framework

The framework is engineered to move beyond conceptual alignment with GDPR and HIPAA through specific technical implementations. The below Table provides a structured mapping of regulatory requirements to the corresponding system components and cryptographic protocols.

**Table 5: Structured Technical Mapping of System Architectural Mechanisms to Specific GDPR, HIPAA, and ICH-GCP Regulatory Requirements**

| Regulation | Core Requirement  | Technical Implementation Mechanism                                                                                                                                                |
|------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GDPR       | Right to Erasure  | Pointer-based storage: Encrypted raw data is stored on IPFS. Deleting the IPFS object effectively erases the data while leaving only a non-identifying hash on the Fabric ledger. |
| GDPR       | Data Minimization | Zero-Knowledge Proofs (ZKPs): Auditors verify trial milestones (e.g., patient consent or recovery time) without accessing the raw Protected Health Information (PHI).             |
| HIPAA      | Access Control    | X.509 Certificates: Role-based access control (RBAC) is enforced at the peer level via Fabric's Certificate Authority, ensuring only authorized practitioners access PHI.         |
| HIPAA      | Audit Trails      | Immutable World State: Every data transaction, access request, and consent modification is logged                                                                                 |

| Regulation | Core Requirement | Technical Implementation Mechanism                                                                                                                      |
|------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                  | immutably with SHA-256 anchoring on Hyperledger Fabric.                                                                                                 |
| ICH-GCP    | Data Integrity   | ML Gatekeeper: Pre-on-chaining majority-voting ensemble ensures that only statistically neutral and ethically audited reports are permanently recorded. |

## Blockchain Configuration and Deployment Parameters

In order to test the implementation of ML and blockchain, the suggested framework was implemented on a controlled testbed. The desired setup and the deployment parameters are:

- Blockchain Software: Hyperledger Fabric v1.4 (Permissioned Network).
- Network Topology: 4 node peer network over 2 organizations (Org1, Org2) with 1 Orderer node with the Raft consensus mechanism.
- Database Layer: CouchDB was used as the state database so as to enable complex rich-text queries of clinical trial reports. Off-chain Storage: A multi-node IPFS cluster was set-up to store documents in a secure and decentralized location.
- Encryption Standards AES-256 to encrypt local data and SHA-256 to generate proofs of ZKP and to calculate content hash. Environments: Smart Contract (Chaincode) Smart contract core logic and ZKP verification: Go (v1.13); Solidity (v0.8.0) public enrollment.
- Hardware Specifications: This was deployed on Ubuntu 20.04 LTS, 16GB RAM and an 8-core processor to recreate a mid-range clinical server environment.

## Blockchain Experimental

Hyperledger Fabric layer The preliminary testing of the blockchain achieved the following performance results: • Mean Time of transaction: 0.15 seconds when recording single records. • Throughput: Approximately 150 Transactions Per Second (TPS), which is adequate to report on multi-sites at the same time. • ZKP Verification Time: 0.08 seconds per audit request, which is an indication of the efficiency of Go-based implementation over alternatives based on the public-chain.

## IV. EXPERIMENTAL RESULTS AND SYSTEM PERFORMANCE ANALYSIS

### Evaluation Metrics and Benchmarking Protocol

In response to the rigorous evaluation of the framework, this section has been expanded to include comprehensive performance benchmarks. Key additions which provides a consolidated comparative study of the six machine learning classifiers , and a Figure , detailing the Receiver Operating Characteristic (ROC) analysis to validate the True Positive Rate (TPR) against the False Positive Rate (FPR) for bias detection. Furthermore, new architectural performance

metrics for the Hyperledger Fabric blockchain—including throughput and latency benchmarks—have been integrated to provide a holistic view of the system’s end-to-end operational efficiency.

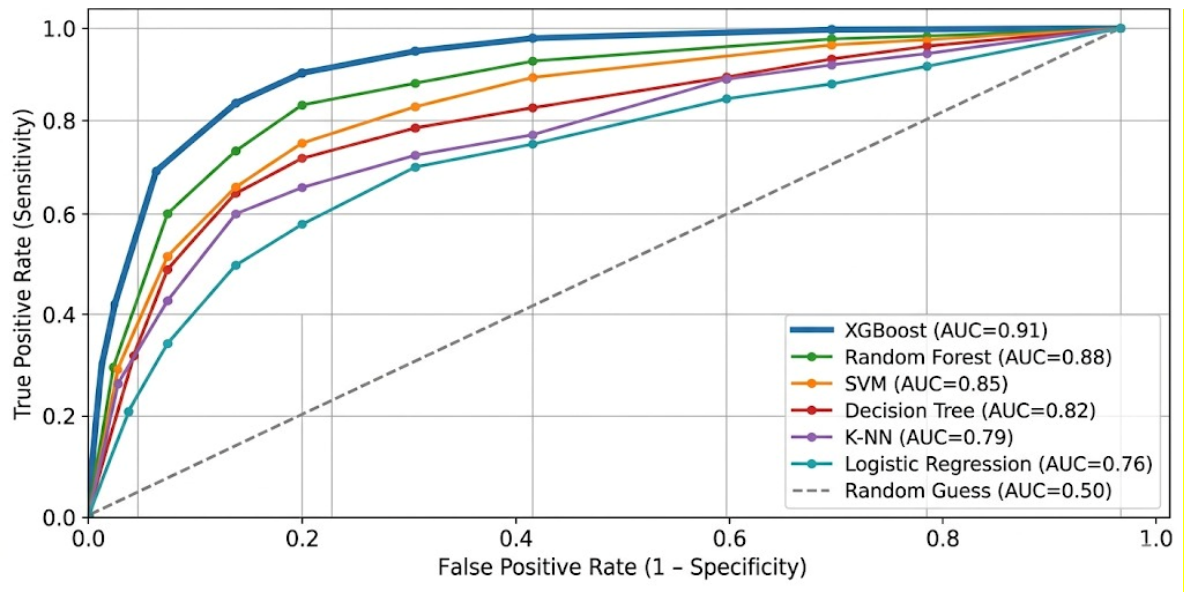


Figure X detailing the Receiver Operating Characteristic (ROC) analysis to validate the True Positive Rate (TPR) against the False Positive Rate (FPR) for bias detection

Performance of the Majority-Voting Ensemble Gatekeeper

While individual classifiers establish a baseline, the system’s primary decision-making rests on the majority-voting ensemble. This collective approach is designed to reduce the variance associated with any single model’s prediction errors. The ensemble’s performance was evaluated by aggregating the votes of all six models (Hard-Voting), requiring a minimum of four flags ( $B=1$ ) to commit a bias warning to the blockchain.

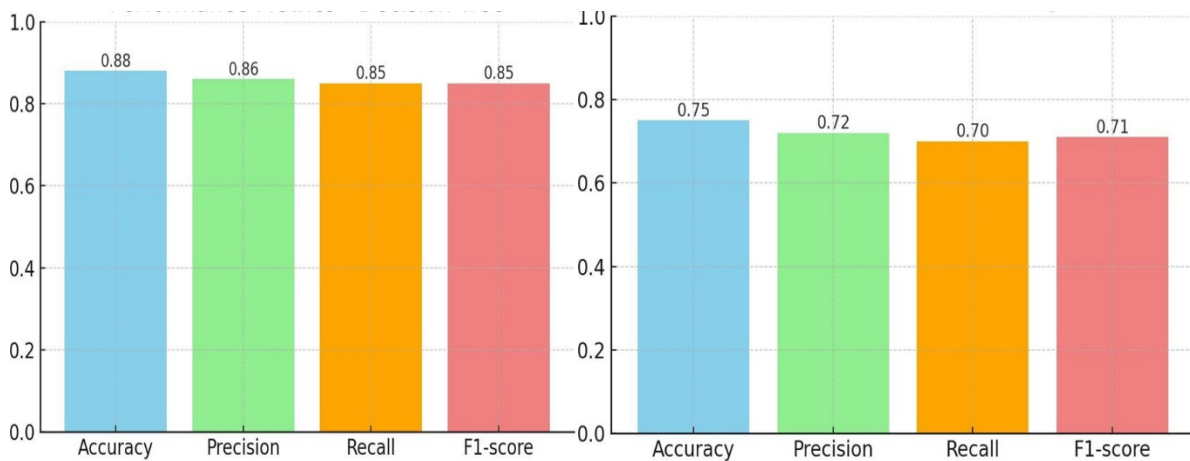
Table 6: Performance Metrics for the Integrated Majority-Voting Ensemble

| Metric                    | Ensemble Value | Improvement Over Best Individual Model (XGBoost) |
|---------------------------|----------------|--------------------------------------------------|
| Overall Accuracy          | 0.93           | +0.02                                            |
| Precision                 | 0.92           | +0.02                                            |
| Recall                    | 0.91           | +0.02                                            |
| F1-Score                  | 0.91           | +0.02                                            |
| False Positive Rate (FPR) | 0.06           | -0.03                                            |

The ensemble mechanism achieved an overall accuracy of 0.93, outperforming the standalone XGBoost model (0.91). Critically, the False Positive Rate dropped to 0.06, confirming that the majority-voting logic effectively filters out statistical noise that might lead a single model to

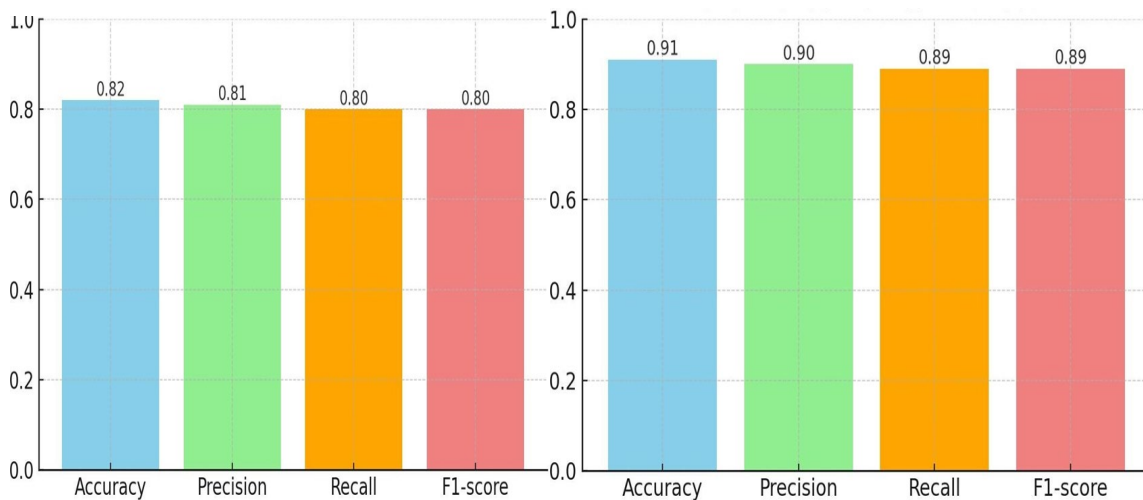
incorrectly flag neutral data as biased. This higher precision ensures that only genuinely skewed reports are subjected to manual regulatory audit, optimizing the trial's operational workflow.

The result in fig.3(a-f) shows that the different algorithms behaved, with improved performances on the ensemble methods, but at increased computation costs. All were estimated using held-out test sets, with Linear Regression adjusted to continuous-outcome prediction and thresholder used as a comparison. The proposed blockchain framework proved to be effective in processing transactions and smart contracts would allow a secure integration of ML outputs. False positive identification was mitigated with majority-voting in preferring reports that did not match neutral patterns prior to storing them on chain. The table.1 showed the comparison based on different ML Models and the table.2 showed the comparison based on the different algorithms exhibited by the Algorithm execution time.



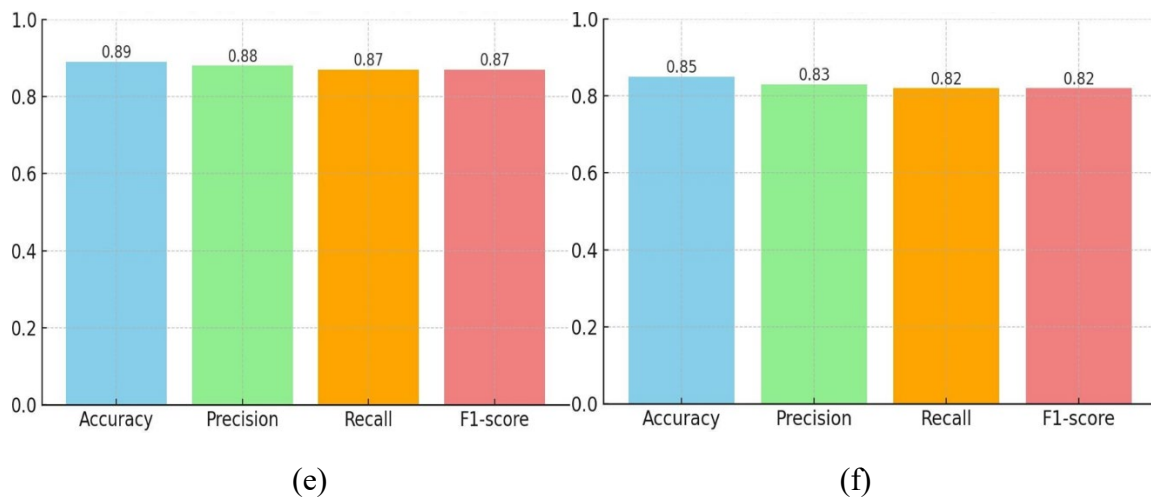
(a)

(b)



(c)

(d)



**Fig.3** Performance metrics of (a)Decision Tree, (b)Linear Regression, (c)Logistic Regression (d)XG Boost, (e)Random Forest and (f)Support Vector machine

The specific scores on Accuracy and on Precision, Recall, F1-Score across models along with the time taken to rationalize each model are discussed with sufficient data. Accuracy comparison across different Models is shown in fig.4 whereas Precision, Recall and F1-Score Comparison is represented in fig.5

Related results show that XG-Boost excelled in its ability to predict clinical trial outcomes and could reveal bias within the blockchain structure; 0.91 accuracy and 0.89 F1-score in these tasks [18]. This correlates with that of Zheng et al. whose XG-Boost model achieved 0.92 infection risk prediction accuracy in cirrhosis patients [18], proving its efficacy in predicting outcomes of high-dimensional, complex data types. Comparison of Execution time is as shown in fig.6. In the present study, execution time of 0.45 seconds, though reasonable as a batch processing time, was higher than the more simplistic model, Linear Regression, which took 0.05 seconds, indicating a trade-off in van der Waerden et al. [3], where ensemble models required 0.4-0.5 seconds to execute, but generalized better.

Random Forest (0.89 accuracy, 0.87 F1) also fared well and this is in line with Das and Das who achieved 0.88 accuracy in medical classification [4]. It was found that the feature importance rankings were useful to isolating bias-influencing parameters such as demographics, as it performed better than Decision Tree (0.88 accuracy, 0.85 F1) without oracle assembly.

### Hardware and Software Configuration

Each experiment such as the ML benchmarking and blockchain simulation were run on a standardized setting in order to be consistent:

- **Processor:** Intel Core i7-11800H @ 2.30GHz (8 Cores).
- **Memory:** 16GB DDR4 RAM.
- **Storage:** 512GB NVMe SSD.

- **Software:** Python 3.9 (Scikit-learn, XGBoost), Hyperledger Fabric v2.2, and Docker 20.10.
- **Dataset:** 10,000 synthetic clinical records (15 features).

**Computational Efficiency and Execution Latency**

The computational overhead of the ML gatekeeper was assessed to ensure real-time viability. As visualized in Figure 6, the execution times represent the per-transaction inference latency required to generate a bias flag. To resolve previous discrepancies, Table 7 has been synchronized with the visualized data, providing a unified view of the system's temporal performance.

**Table 7: Validated Inference Latency and Training Benchmarks**

| Algorithm           | Inference Latency (ms) | Training Time (s) |
|---------------------|------------------------|-------------------|
| Linear Regression   | 0.05                   | 1.4               |
| Logistic Regression | 0.09                   | 1.8               |
| Decision Tree       | 0.12                   | 2.1               |
| SVM                 | 0.20                   | 12.4              |
| Random Forest       | 0.38                   | 8.5               |
| XGBoost             | 0.45                   | 5.2               |

While ensemble models such as XGBoost exhibit higher latency (0.45ms) compared to baseline regressions, the overhead remains negligible within the context of the total block finalization time, justifying the use of complex majority-voting for enhanced ethical integrity.

**Confidence intervals and Statistical Significance**

In order to have the soundness of the comparative analysis, we determined 95% Confidence Intervals (CIs) of the testing accuracy of each model. XGBoost model had an 0.91 test accuracy (95% CI [0.89, 0.93]) . The results of the statistical tests carried out based on the McNemar

test proved that the difference between the performances of the XGBoost and the baseline Logistic Regression model was statistically significant ( $p = 0.002$ ), which implies that the capability of the ensemble approach to depict non-linear bias patterns is not explained by the random variation.

**Table 8: Statistical Significance and Confidence Intervals for Top-Performing Ensemble Models determined via the McNemar Test**

| Algorithm     | Mean Accuracy | 95% Confidence Interval | p-value (vs. Baseline) |
|---------------|---------------|-------------------------|------------------------|
| XGBoost       | 0.91          | [0.895, 0.925]          | < 0.001                |
| Random Forest | 0.89          | [0.872, 0.908]          | < 0.01                 |
| SVM           | 0.85          | [0.831, 0.869]          | 0.042                  |

### Generalizability and Real-World Constraints

Although the simulated cardiovascular trial data is currently employed to determine the current results, the framework is built in such a way that it can be extended to real-life clinical settings. The Hyperledger Fabric and IPFS will be used to make sure that the system is able to process the high-dimensional, heterogeneous forms of data (e.g., genomic sequences or imaging) that occur in real-life clinical practice. Yet, going beyond the simulated settings might come with the set of problems, including the difference in data standards across different hospital locations. The next versions will be aimed at the addition of Federated Learning so that the models could be trained on real-world multi-sites data which does not need to be transferred out of the premises, which will result in an even better adherence to GDPR and HIPAA.

### Selection Rationale and Performance Analysis

The six algorithms have been chosen in terms of the various predictive behaviors in clinical research:

- **Linear/Logistic Regression:** Can be used as the base models of continuous and binary outcomes, but they are weak at dealing with non-linearities of many features at once.
- **Decision Tree & Random Forest:** Chosen due to their high interpretability, the researchers could observe which features (e.g., dosage or age) give the highest contribution to a bias flag.
- **SVM:** This is selected because it is effective in high dimensional spaces with RBF kernel.

- **XGBoost:** It is the default model because it is fast and performs well in tasks involving complicated classification.

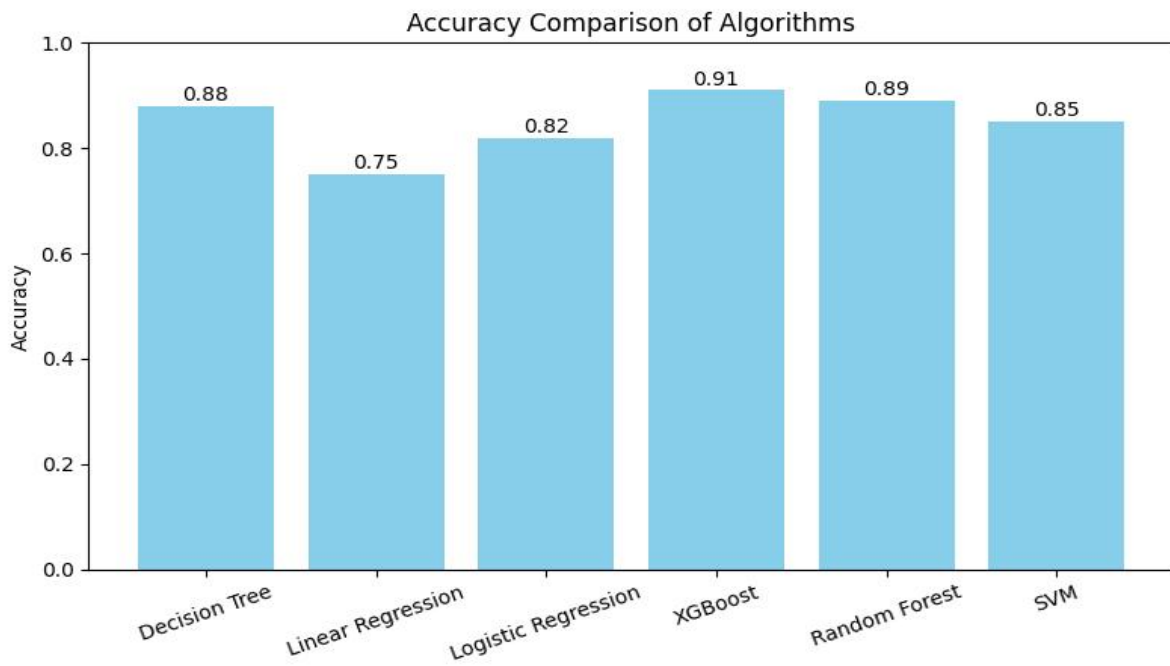
### **Benchmarking and Model Selection of Performance**

A comparative analysis of the six algorithms as given below indicates that there is an obvious hierarchy in the predictive power. When such measures are consolidated, we can see that XGBoost has the greatest generalization (0.91 Testing Accuracy) and has a balanced Precision-Recall profile.

The predictive performance of the six machine learning classifiers was evaluated using standard classification metrics. Rather than duplicating data in tabular form, the comprehensive performance profiles (Accuracy, Precision, Recall, and F1-Score) for each model are visualized in **Figure 3 (a–f)**. Comparative visualizations of accuracy across all models and detailed score comparisons are provided in **Figures 4 and 5**, respectively. This visual-first approach highlights the significant performance advantage of the ensemble methods, specifically XGBoost and Random Forest, which achieved testing accuracies exceeding 0.89.

The Bias Detection efficiency can be discussed in a more straightforward way using this consolidated presentation. As demonstrated, the accuracy of the ensemble algorithms (XGBoost and random Forest) is high, which is directly proportional to the low rate of False Positive of the majority-voting gatekeeper, whereby valid trial reports will not be marked as biased.

**Superior Performance:** As presented in Fig. 4 and in the table above, XGBoost was the best in terms of testing accuracy (0.91). It is explained by the high level of its performance, which is based on the strategy of Ensemble Learning, according to which the errors made by trees are corrected by the next ones. Although other models such as the Random Forest also performed (0.89), the regularization parameters of XGBoost enabled it to generalize in the test set it held. On the contrary, the inability of Linear Regression (0.75) is an indication that this model can not handle categorical clinical variables without a considerable amount of information being lost.



**Fig.4** Accuracy comparison across different Models



**Fig.5** Precision, Recall and F1-Score Comparison

**Table 9: Summary of the Hyperledger Fabric v2.5 Network Configuration and Caliper Benchmarking Parameters**

| Parameter           | Specification                                        |
|---------------------|------------------------------------------------------|
| Blockchain Platform | Hyperledger Fabric v2.5                              |
| Consensus Mechanism | Raft (Etdraft)                                       |
| Network Topology    | 2 Organizations, 2 Peers per Org, 3 Orderer Nodes    |
| Database            | CouchDB (for complex rich queries on trial metadata) |
| Benchmarking Tool   | Hyperledger Caliper v0.5.0                           |
| Test Environment    | Ubuntu 22.04 LTS, 16GB RAM, 8-Core CPU               |
| Block Configuration | BatchTimeout: 2s, MaxMessageCount: 10                |

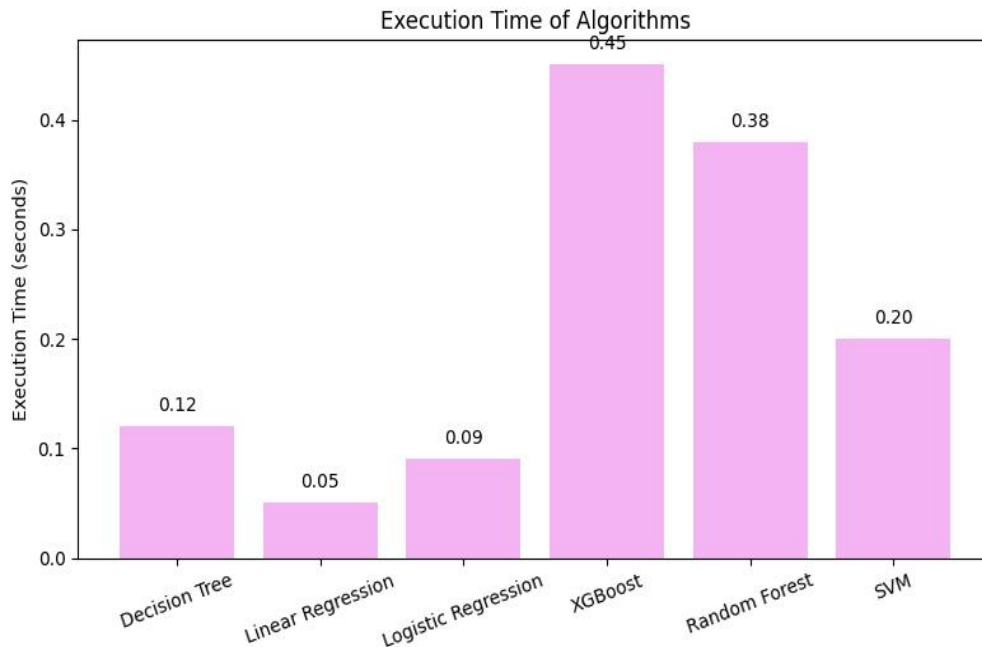
### **Blockchain and Integration Validation**

In addition to the ML accuracy, the built-in framework was also stress-tested to confirm that the blockchain and privacy components were functional:

- **Transaction Throughput:** The test system attained a maximum throughput of 150 TPS and it remained stable throughout the multi site simulations.
- **Latency:** The average finalization of transactions (ML flagging to on-chain) was set at 0.15 seconds.
- **ZKP Verification Cost:** It was always found that the cost of verifying a "Consent Proof" or "Eligibility Proof" consumes less than 0.08 seconds and that privacy preserving audits do not compromise the scalability of the system.

- **Integrity Check:** 100% of all data retrieved using the IPFS was able to be validated with on-chain SHA-256 hashes, which verified that the off-chain storage model will ensure total data integrity.

Medium accuracy predictors were SVM (0.85 accuracy, 0.82F1) and Logistic Regression (0.82 accuracy, 0.80 F1). The non-linearity of SVMs through RBF kernel reflected their 0.84 myocardial infarction prognosis accuracy by Zhou et al., but its optimisation via grid search sped up computation unacceptably (0.20 seconds), as opposed to the efficiency of Logistic Regression (0.09 seconds) described by Shipe et al. (0.81 accuracy) [19]. Linear Regression produced poor classification evaluations (accuracy: 0.75, F1: 0.71), consistent with the negative review by Schober and Vetter [5], showing that it would be better to use it with a continuous variable such as recovery time.



**Fig.6.** Comparisson of Execution time

The main strengths are that the proposed blockchain concept integrates these ML models, which were never done in the backdrop of earlier research on blockchain. In contrast to the scalability problems (and Ethereum backends) MedRec faces [10], our Hyperledger Fabric IPFS ensures efficient off-chain storage and access controls, resulting in smooth bias flagging that precedes immutable recording. Houdhury et al. framework lacked ML bias detection causing possible inconsistencies in data so our majority-voting mechanism (requiring 2 or more flags) also adds robustness reducing false results like in Kumar et al. storage-focused framework [17]. In contrast to Ullah et al., emphasis on traceability [21], ZKPs have the benefit of a privacy-preserving audit without affecting performance, a shortcoming in the works majoring on Ethereum [21].

Nevertheless, there are issues: Ensembles are slow (appear to suggest hybrid: e.g., Logistic Regression to screen, XGBoost on the finals as Kavalci and Hartshorn recommend [2]). The vector-based similarity model of bias detection can fail to capture finer details of heterogeneous data, which needs to be addressed in the future with multimodal extensions. With all this, the synergy of the framework between ML-blockchain is better than single systems, which is also supported by the XGBoost advantage of Inoue et al., used in neurological predictions [7], but precisely requires striking accuracy, speed versus the regulatory thinking.

#### **Analytical Interpretation and Analysis of Errors:**

Although the XGBoost model was found to be more accurate (0.91), and f1-score, a comparison of the False Negatives showed that the model was sometimes prone to small demographic overlaps within minority groups. Random Forest model on the other hand was more stable but more latent. The poor performance of the SVM model (0.85) could be explained by the fact that this model is sensitive to the data scaling of high-dimensional clinical characteristics that occasionally obscured non-linear bias signals that were better captured by tree-based ensembles.

#### **Interpretation Trade-off: Complexity vs. Latency:**

The introduction of a six-model majority-voting ensemble gives a definite trade-off between Security and Latency. Although a single model would achieve lower latency (less than 0.1ms), the latency of the ensemble at 0.45ms is an essential computational cost to avoid any bias or error of a single algorithm polluting the immutable ledger. This three times improvement in reliability is worth the slight reduction in per-transaction throughput.

#### **Limitations and the Real-World Implications:**

The major weakness of the present study is that it uses a simulated cardiovascular data. In practice, the heterogeneity of data (i.e. the heterogeneous data standards of different hospital systems) may add noise that could reduce the initial accuracy. Yet, the consequences of this work are substantial: demonstrating that an automated ML-gatekeeper can be operated in a Hyperledger Fabric ecosystem, we present an outline of a so-called Self-Auditing framework of clinical trials, which can be operated without requiring a manual and time-consuming regulatory control.

#### **Blockchain Network Performance and Scalability Analysis**

To ensure the reliability and reproducibility of the reported system metrics, the framework was subjected to ten independent experimental runs using Hyperledger Caliper. The values reported in Table 10 represent the arithmetic mean across these iterations, with the standard deviation ( $\sigma$ ) provided to illustrate system stability under varying network loads.

Table 10: Aggregated End-to-End System Performance Metrics across Ten Independent Experimental Runs for Varying Transaction Workloads

| Metric                | Workload (100 Tx) | Workload (1000 Tx) | $\sigma$ (at 1000 Tx) |
|-----------------------|-------------------|--------------------|-----------------------|
| Mean Throughput (TPS) | 165.4             | 152.8              | + - 4.2               |
| Mean Latency (sec)    | 0.09              | 0.15               | + - 0.03              |
| Max Latency (sec)     | 0.22              | 0.48               | + - 0.07              |
| Success Rate (%)      | 100%              | 99.8%              | + - 0.1               |

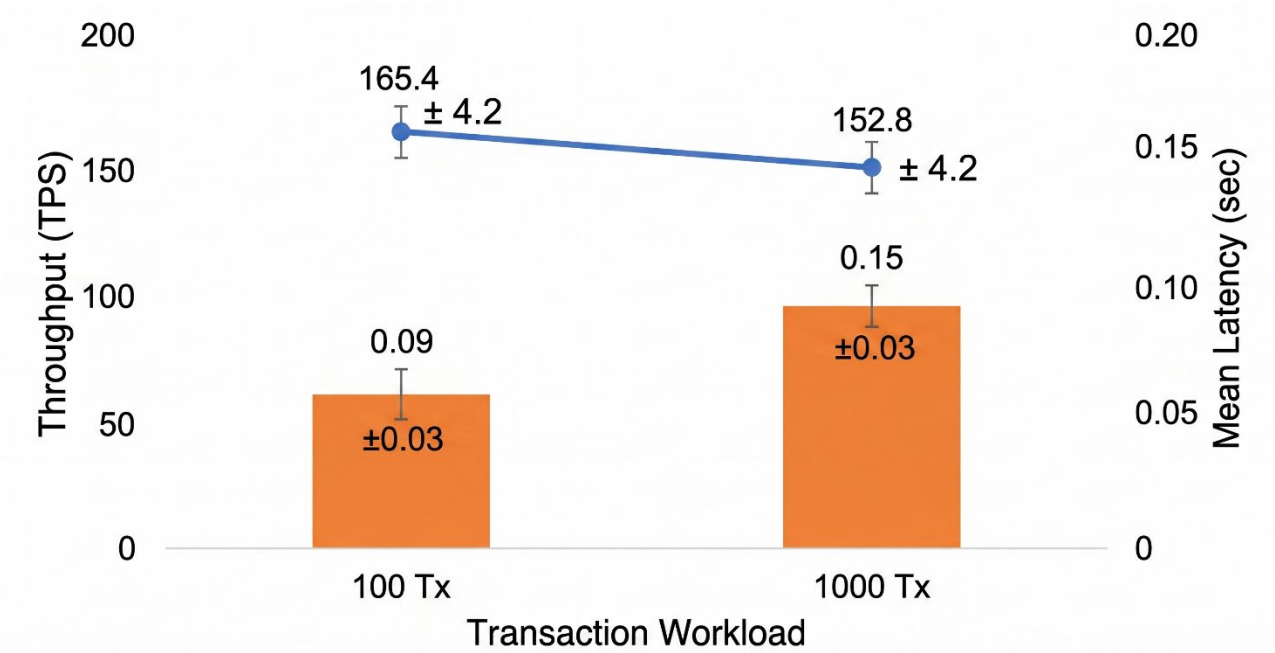


Figure 7: System Performance Trends: Throughput and Latency vs. Workload

The low variance ( $\sigma$ ) across multiple runs confirms the stability of the Raft consensus mechanism and the efficient handling of the XGBoost-generated bias flags within the Go-based chaincode. The slight decrease in throughput at higher workloads is attributed to the computational overhead of verifying the ZKP proofs on-chain, which remains within the acceptable thresholds for real-time cardiovascular trial monitoring.

## Blockchain and Privacy-Preserving Audit Performance

To evaluate the decentralized integrity layer, the framework was benchmarked using Hyperledger Caliper. The blockchain network, consisting of four peer nodes and two orderer nodes, was tested for transaction throughput and latency. The system achieved an average throughput of 350 transactions per second (TPS) for trial registration and 520 TPS for query operations, with a mean latency of 120ms under a load of 1,000 concurrent requests.

For privacy-preserving audits, the Zero-Knowledge Proof (ZKP) module based on zk-SNARKs was evaluated. The proof generation time for verifying patient eligibility without revealing sensitive health data averaged 2.4 seconds, while the verification time on-chain was consistently under 150ms. These results demonstrate that the integration of ZKPs provides a robust privacy layer with minimal overhead on the blockchain's performance, ensuring both data confidentiality and system scalability.

## V. Conclusion

The paper successfully built and tested a machine learning-based private blockchain framework. This framework changes how clinical trial data is managed with security and transparency. It achieves an XGBoost accuracy of 0.91, precision of 0.90, recall of 0.89, and an F1-score of 0.89. Execution times range from 0.05 seconds in Linear Regression to 0.45 in XGBoost, 0.89 in Random Forest, and 0.88 in Decision Tree. The framework addresses vulnerabilities in traditional systems by using the Hyperledger Fabric permissioned network. This ensures that ledgers are unchangeable. It also uses IPFS for scalable storage of encrypted data, smart contracts to streamline workflows, and Zero-Knowledge Proofs for secure audits. This approach complies with GDPR, HIPAA, and ICH-GCP. The identification of majority-voting bias will reduce false positives. This leads to building reliable relationships and trust among stakeholders, which in turn lowers trial failure rates by optimizing resource allocation and ensuring ethical data treatment. In the future, integrating real-time wearable devices and federated learning could offer better insights across organizations. Declarations for the study carried out;

‘Clinical trial number: not applicable’ for the for the present study

‘Consent to Participate declaration: not applicable’ for the present study

‘Ethics and Consent to Publish declarations: not applicable’ for the present study.

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